

## Short communication

Response of plant diversity to moss, *Carex* or *Scirpus* revegetation strategies of wet depressions in restored fens

Bérenger Bourgeois, Line Rochefort\*, Vicky Bérubé, Monique Poulin

Peatland Ecology Research Group, Department of Plant Sciences, Université Laval, Quebec City, Quebec, G1V 0A6, Canada

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## ABSTRACT

Pool margins and wet depressions (hollows) contribute considerably to peatland biodiversity by sheltering specific plant assemblages. In peatlands restored after peat extraction, the typical communities associated with pools generally fail to re-establish if only the moss layer transfer technique is applied, a common approach on flat bare peatlands. To adapt peatland ecological restoration methods for recolonizing shallow wet fen habitats, this study investigates the response of plant communities re-establishing in wet depressions subjected to four revegetation strategies. In a restored fen (southeastern Quebec, Canada), 48 depressions were dug out and revegetated with 1) mosses, 2) *Carex* species, 3) *Scirpus validus*, or 4) through spontaneous colonization (control treatment without implanted species). After four growing seasons, plant community structure and composition were surveyed in and outside the revegetated area respectively to test whether reintroduced communities had a propensity to extend beyond the central wet depression. Restoration strategies strongly impacted plant communities. Moss-revegetated wet depressions had higher moss and vascular fen-specialist species richness and distinct species assemblages relative to other treatments. *Carex* treatment increased only vascular cover compared to control, while *Scirpus validus* treatment did not differ from control. The positive impact of moss transfer was local and limited to core revegetated areas, as transferred plant material did not colonize any further. Our results suggest that transferring moss in created wet depressions is an effective strategy to foster the recolonization of diverse and distinct plant assemblages, and thereby restore local fen biodiversity hotspots.

## 1. Introduction

Peatlands are biodiverse ecosystems that play an important role in soil carbon storage worldwide (Rochefort et al., 2012; Rydin and Jeglum, 2013; Bonn et al., 2016). In these systems, microhabitat heterogeneity strongly drives species diversity, since specific environmental conditions generally promote the establishment of locally rare species. While wooded peatlands encompass more rare bryophytes than open peatlands, canopy gaps have also been found to favor rare vascular species (Johnson and Leopold, 1994; Locky and Bayley, 2006). In particular, peatland pools and wet hollows contribute substantially to plant richness by sheltering distinct species assemblages (Poulin et al., 2002; Whitehouse and Bayley, 2005; Fontaine et al., 2007). Nevertheless, peatlands have experienced

extensive anthropogenic degradation, notably due to conversion to agriculture or forestry, fuel production and peat extraction for horticultural purposes (Joosten and Clarke, 2002; Chapman et al., 2003). In the latter case, once peat extraction activities cease, peatland plant communities generally fail to spontaneously re-assemble (Poulin et al., 2005; Graf et al., 2008) because of altered hydrology (i.e., peatland surfaces are drained and homogenized; Price, 1996), a depleted seed bank (Salonen, 1987; Triisberg et al., 2011), frost heaving (Groeneveld and Rochefort, 2005) and soil erosion (Campbell et al., 2002). Active restoration strategies are thus essential.

In ombrotrophic peatlands (*bogs*), the Moss Layer Transfer Technique (MLTT) has been demonstrated as a successful restoration approach to promote the recolonization of moss carpet. This technique

\* Corresponding author at: Peatland Ecology Research Group, Département de Phytologie, Faculté des Sciences de l'Agriculture et de l'Alimentation, Université Laval, 2425 rue de l'Agriculture, Québec, Québec, G1V 0A6, Canada.

E-mail addresses: [berenger.bourgeois.1@ulaval.ca](mailto:berenger.bourgeois.1@ulaval.ca) (B. Bourgeois), [line.rochefort@fsaa.ulaval.ca](mailto:line.rochefort@fsaa.ulaval.ca) (L. Rochefort), [vicky.berube.1@ulaval.ca](mailto:vicky.berube.1@ulaval.ca) (V. Bérubé), [monique.poulin@fsaa.ulaval.ca](mailto:monique.poulin@fsaa.ulaval.ca) (M. Poulin).

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consists of (i) re-establishing hydrological conditions through ditch blocking and soil profiling, (ii) spreading moss diaspore and living material obtained from the upper layer of a natural peatland at a 1:10 ratio, (iii) protecting moss diaspore with straw mulch, and eventually (iv) fertilizing with phosphorous (Rocheffort et al., 2003). In these restored bogs, typical peatland plant communities generally recolonize within five years (González et al., 2013). Despite the well-recognized success of bog restoration (Chimner et al., 2017), effective strategies in the case of minerotrophic peatlands (*fens*) are still under development (Lamers et al., 2015; Rocheffort et al., 2016). In Colorado and Alberta, innovative fen restoration and creation projects have, however, met with success by planting willows and *Carex* species, combined with rewetting by ditch blocking (Cooper and MacDonald, 2000; Vitt et al., 2011; Cooper et al., 2017). Recently, mesocosm experiments have also shown that conditions for fen initiation can be recreated after oil sands mining by introducing foundation moss and a mix of vascular species (Borkenhagen and Cooper, 2016).

Plant communities specific to bog pool margins are generally recalcitrant to MLTT restoration strategies. While moss transfer at pool margins promotes good establishment of *Sphagnum* mosses after two growing seasons (up to 60% cover; Poulin et al., 2011), which subsequently reduces germination of the invasive species *Typha latifolia* (Bourgeois et al., 2012), re-establishment of vascular plant species remains minimal (below 5% cover; Poulin et al., 2011). Additional cost-effective techniques must thus be implemented to restore these local biodiversity hotspots. In particular, field experiments have shown that establishing ericaceous shrubs along pool margins of restored bogs stabilizes peat and fosters *Sphagnum majus* establishment (Laberge et al., 2013). The presence of *Sphagnum* carpets further promotes the growth of *Carex* introduced as seeds (*C. limosa*, *C. magellanica*, *C. oligosperma*; Laberge et al., 2015). Strategies for restoring entire plant communities along pool margins and wet depressions of fens remain to be developed, however.

This study aims to assess the effect of vegetation transfer strategies on plant communities establishing in shallow fen pools. More precisely, we compared plant communities of wet depressions created in a restored fen in eastern Canada and subjected to four plant reintroduction treatments (i.e. mosses, *Carex* or *Scirpus* transfer, and no vegetation transfer as a control). We hypothesized that 1) both *Carex* and *Scirpus* transplants establish successfully and support high vegetation cover and 2) moss transfer promotes higher fen-specialist plant richness and distinct species assemblages. The underlying ecological mechanisms of plant community re-establishment are then discussed, together with practical implications for future restoration projects.

## 2. Materials and methods

### 2.1. Study site

This study was conducted at the Bic-Saint-Fabien research station (48°19' N, 68°50' W), a peatland complex located in the Lower Saint-Lawrence region (SE Québec, Canada), characterized by a cool sub-humid climate (average annual temperature: 4.4 °C; average annual precipitation: 958.5 mm, of which ca. 30% falls as snow; climate-weather.gc.ca). The station includes 6 ha of natural open moderate-rich fen, 10 ha of natural wooded moderate-rich fen, and 15 ha of post-extracted peat sectors. Peat extraction for horticultural purposes took place from 1946 to 2000, or until the bog peat was completely removed in some sectors. The remaining peat, mostly composed of layers of brown moss stems and sedges, corresponded to moderately-rich fen conditions and varied from a few cm to 1–2 m thick (Bérubé et al., 2017). The water table in the cutover sectors was on average –21.5 cm below the surface prior to restoration in 2009, but was raised to 8.6 cm in 2011 (Malloy and Price, 2014). Analyses from July 2009 showed that the surface pore water had a mean pH of 6.2

( $\pm 0.6$ , standard deviation;  $n = 19$ ) and a mean corrected electric conductivity of  $338 \mu\text{S}\cdot\text{cm}^{-1}$  ( $\pm 247$ , standard deviation;  $n = 19$ ; Rocheffort, unpublished results). Post-extracted peat sectors were characterized by bare peat (61% cover on average), and a spontaneous vegetation dominated by *Calamagrostis canadensis* (21%), *Typha latifolia* (15%), *Equisetum arvense* (13%), *Scirpus cyperinus* (11%), and *Spiraea alba* var. *latifolia* (10%).

The restoration took place from late fall 2009 to early summer 2010 and is described in detail in Rocheffort et al. (2016). Briefly, over 2.2 ha were restored following the MLTT. Moss material, extracted in May 2010 from a nearby moderately-rich riparian fen dominated by *Sarmenthyphnum exannulatum*, *Campylium stellatum* and *Trichophorum alpinum*, was spread mechanically in a 1:10 ratio (donor surface:restored surface) on bare peat surfaces and protected with straw. Two other sectors were restored by either 1) profiling only the surfaces, to level the dome shape typical of residual peat fields and eliminate spontaneous vegetation (over 3.6 ha) or 2) rewetting without profiling to preserve the spontaneously established typical fen vegetation (4.9 ha). The former drainage system was blocked in June 2010, leading to successful rewetting of the site (Malloy and Price, 2014). After six growing seasons, vegetation re-establishment was lower than expected for all sectors, but the profiling technique led to the lowest rate of recolonization by typical fen species (Rocheffort et al., 2016). Active plant re-introduction was therefore highly recommended to restore typical plant communities, but research was still required to develop adequate techniques.

### 2.2. Experimental design

In spring 2010, 48 shallow bowl-shaped depressions (or wet hollows, corresponding to 48 experimental units) of  $2.5 \times 3.5$  m, covering a water depth gradient ranging from 0 to 80 cm, were dug mechanically in the re-profiled sectors. The wet depressions were then weeded manually when necessary and subjected to four restoration treatments following a completely randomized block design: 1) moss layer transfer, 2) *Carex* transplantations, 3) *Scirpus validus* transplantations, and 4) no plant introduction (control). More precisely, treatments consisted of 1) spreading moss mixtures dominated by either *Campylium stellatum* for the drier three-fourths at the periphery of each created depression, or by *Scorpidium scorpioides* for the central deeper last fourth, 2) the plantation of four *Carex* species at a density of 42 individuals·m<sup>-2</sup> (one individual every 15 cm), corresponding to 138 *Carex aquatilis*, 138 *Carex utriculata*, 46 *Carex lasiocarpa*, and 46 *Carex lacustris* per plot, 3) the plantation of a 10 cm long rhizome section of *Scirpus validus* (every 20 cm for a total of 204 rhizome sections), or 4) spontaneous recolonization without revegetation (control). The reintroduction treatments were applied in May 2010 to the entire depression surface, including the deeper center. Due to the rarity of natural fens in the areas surrounding the study site, plant material for moss transfer was collected in a moderately rich fen on the northern shore of the Saint Lawrence River, 400 km northeast of the site. The upper layer of vegetation (3–5 cm, including mosses) was cut, transported to the site, and spread at a 1:5 ratio (1 m<sup>2</sup> of collected material spread over 5 m<sup>2</sup> of reintroduction surface – a higher ratio than recommended, to accelerate moss colonization). No phosphorus fertilization was applied due to the proximity of water surfaces; straw mulch was spread on mosses. Plugs and rhizomes of vascular species (*Carex* spp. and *Scirpus validus*) were collected in a drained riparian fen, 15 km west of the research site.

Plant communities were surveyed after four growing seasons (August 2013). Moss and vascular species were inventoried in four circular plots of 70 cm diameter placed inside each created depression, 50 cm from the edge of the plots. The potential effect of vegetation transfer on the plant community established just outside the reintroduction zone was assessed in six supplementary circular plots

of 70 cm diameter placed 15 cm outside each wet depression. In each plot, plant community structure was visually estimated based on the cover of two vegetation strata (i.e. moss and vascular species), and plant community composition based on the cover of each species (in %). Relying on expert knowledge, the 61 spontaneous plant species inventoried were further classified into fen-specialists (four mosses and 40 vascular species) and generalists (no moss and 11 vascular species); six taxa identified at the genus level remained unclassified (Appendix A). The cover of strata and species in the four plots placed within wet depressions and in the six plots placed around them was then averaged to obtain two composite plant surveys per unit (i.e. one within the reintroduction zone and one surrounding the reintroduction zone, hereafter referred to as within and around wet depressions, respectively).

### 2.3. Statistical analyses

The response of plant communities to restoration strategies after four growing seasons was assessed based on three vegetation metrics, namely vegetation structure (moss and vascular plant cover, including transferred individuals due to the difficulty of disentangling planted and spontaneous individuals), species richness and species composition (excluding implanted species). First, the effect of treatments on (i) the cover of mosses and (ii) the cover of vascular species was quantified using ANOVA, followed by Tukey post-hoc tests. A similar procedure was used to identify the effect of treatments on (i) the richness of mosses, and (ii) the richness of vascular species, both for fen-specialists and generalists. Finally, species composition between treatments was compared using a Redundancy Analysis on Hellinger-transformed species cover. Analyses were conducted in R version 3.3.3 (R Core Team, 2017).

## 3. Results

### 3.1. Effect of revegetation strategies on plant community structure

Plant community structure within wet depressions was strongly affected by restoration strategies after four growing seasons. Treatments indeed influenced moss ( $F = 19.87$ ;  $p < 0.0001$ ; adjusted- $R^2 = 54.64\%$ ) and vascular cover ( $F = 11.23$ ;  $p < 0.0001$ ; adjusted- $R^2 = 39.50\%$ ). More precisely, moss cover was close to 30% within wet depressions revegetated with mosses, while almost null in other treatments. Only *Carex* plantation led to higher vascular species cover within wet depressions relative to the other treatments, with, in particular, a cover of vascular species 1.7 times higher on average (min.: 0.2 – max.: 16.0) compared to moss transfer (Fig. 1). Overall, unlike mosses and *Carex*, *Scirpus validus* failed to establish after four growing seasons, and hence did not impact vegetation structure within wet depressions compared to control.

### 3.2. Effect of revegetation strategies on species richness

The richness of fen-specialist plant species within wet depressions was affected by restoration strategies both for moss ( $F = 10.43$ ;  $p < 0.0001$ ; adjusted- $R^2 = 37.58\%$ ) and vascular species ( $F = 29.94$ ;  $p < 0.0001$ ; adjusted- $R^2 = 64.87\%$ ; Fig. 1). In fact, moss transfer led to an increase of fen-specialist moss and vascular species richness (up to 3 moss and 17 vascular species) compared to the other treatments, though moss richness remained low for all treatments. When only generalist vascular species are considered (see Appendix A), richness was similar between restoration strategies ( $F = 0.90$ ;  $p = 0.4502$ ) but very low, with  $1.29 \pm 0.14$  generalist vascular species observed per wet depression on average (mean  $\pm$  standard error; data not shown).

### 3.3. Effect of revegetation strategies on plant community composition

Restoration strategies had a strong influence on the composition of

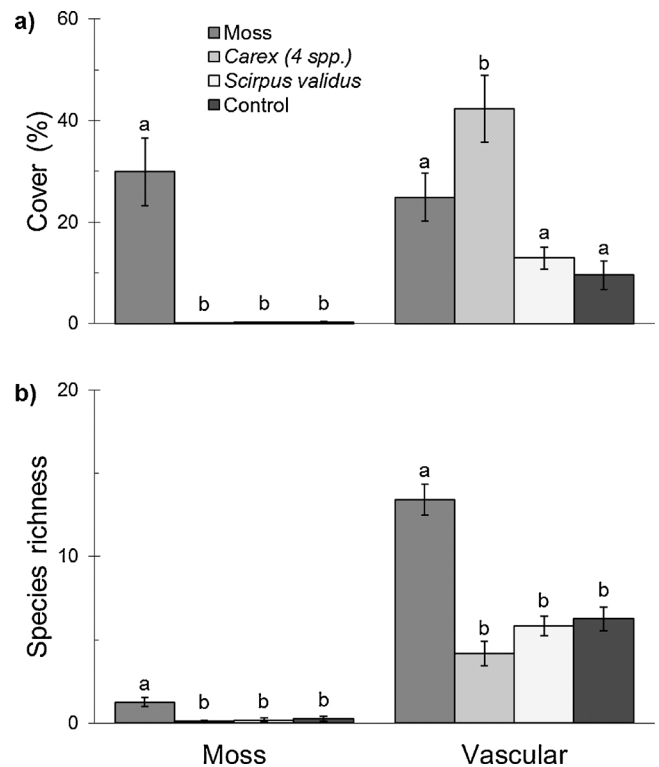
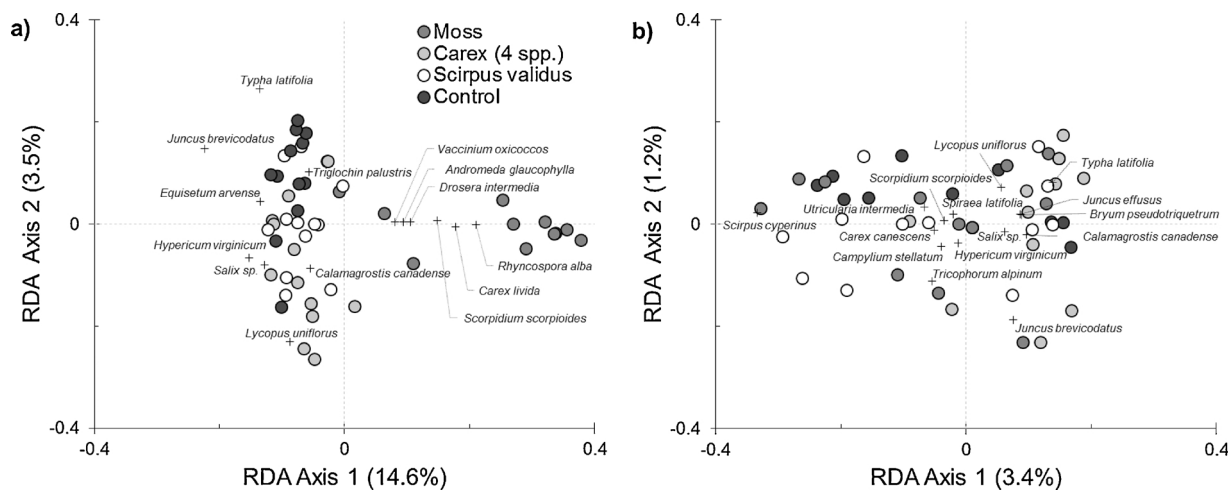


Fig. 1. a) Plant community structure (i.e. moss and vascular cover), and b) richness of moss and vascular fen species (excluding implanted species), within depressions for each vegetation transfer treatment (mean  $\pm$  standard error). Letters refers to the treatment groups, obtained by Tukey post-hoc tests.

plant communities within wet depressions ( $F = 3.60$ ;  $p < 0.0001$ ). Species assemblages within moss-revegetated wet depressions were characterized by high covers of *Rhynchospora alba*, *Carex livida*, *Drosera intermedia*, *Andromeda glaucophylla* and *Vaccinium oxycoccos*, and clearly differed from those observed for the other treatments, as shown by the distinct separation of these two groups of plant communities along the first axis of Redundancy Analysis (Fig. 2; Appendix A). In addition, species assemblages within wet depressions subjected to multiple *Carex* species or a *Scirpus validus* transfer differed slightly from the control treatment along the second canonical axis. Conversely, around wet depressions, community composition was not impacted by restoration strategies, and indeed was similar regardless of the treatment applied ( $F = 0.03$ ;  $p = 0.6591$ ; Fig. 2). In short, revegetation strategies only facilitated spontaneous plant establishment inside the immediate zone of reintroduction.

## 4. Discussion

After four growing seasons, the recolonization of plant communities along margins of created wet depressions in restored fens is largely conditioned by the nature of the restoration strategy conducted. *Carex* reintroduction fostered higher vascular cover within created wet depressions, which may support the stabilization of the peat substrate despite periodically flooded conditions, but without significant impact on fen-specialist richness and plant composition. Conversely, *Scirpus* reintroduction did not modify structure, richness or composition compared to the control treatment. Moss transfer led to different community structure and composition within wet depressions, characterized by higher moss cover and higher fen-specialist richness recruitment than the three other treatments. The positive response of plant communities to moss transfer was, however, very local and limited to the revegetated area of the wet



**Fig. 2.** Composition of plant communities a) within and b) around wet depressions subjected to four revegetation strategies, obtained by Redundancy Analysis on Hellinger-transformed species abundance. Only community composition within wet depressions was significantly influenced by revegetation strategies. The 15 species contributing the most to each ordination space are indicated.

depressions. Around wet depressions subjected solely to spontaneous colonization, the influence of revegetation strategies on plant composition was indeed absent.

Several ecological mechanisms can be proposed to explain the contrasting response of plant communities to vegetation transfer strategies. First, the presence of plant propagules in the moss material transferred to the wet depressions must have promoted the increase of species richness through diaspore dispersal. Furthermore, the moss carpets established after vegetation transfer may have provided suitable seedbeds for the spontaneous establishment of distinct species assemblages by facilitation effect (Groeneveld et al., 2007; Laberge et al., 2015), establishment limitation of invasive species like *Typha latifolia* (see Appendix A; Bourgeois et al., 2012), and higher soil stability (i.e. reduced wind erosion and frost heaving; Groeneveld and Rochefort, 2005; Laberge et al., 2013). In wet depressions subjected to a *Carex* or *Scirpus* revegetation strategy, in contrast, competition for light with implanted species (Kotowski et al., 2006) and low immigration potential of vascular species (Campbell et al., 2003) are likely to have prevented recolonization of fen-specialist species.

**Implications for management:** the moss layer transfer approach should be favored for the revegetation of wet depressions in future restoration projects of post-extracted fens in order to promote both the diversification of plant communities and the recolonization of

adapted fen plant assemblages. Long-term monitoring remains advisable, however, to assess the trajectory and stability of restored communities after four growing seasons. Care should also be taken to select an appropriate pool margin microhabitat, as close as possible to the restored site, from which to collect plant material for transfer. The reintroduction of a mix of *Carex* species by seedling transplantation has been shown to be a good approach to revegetate and stabilize the bare peat substrate, which can remain bare for decades without active reintroduction (Cooper and MacDonald, 2000). Finally, future studies should consider testing the introduction of any species mix, in particular for approaches combining moss transfer and *Carex* transplants, which could foster plant diversity, achieve good coverage of the peat surfaces and prevent invasive species establishment as well as peat erosion and degradation.

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Appendix A. Vegetation strata and plant species cover per treatment

Table A1

**Table A1**  
Average cover of vegetation strata and plant species per treatment (in %). F and G superscripts respectively indicate fen-specialist and generalist spontaneous species.

|                   |                        | Within hollow |       |         |         | Around hollow |       |         |         |
|-------------------|------------------------|---------------|-------|---------|---------|---------------|-------|---------|---------|
|                   |                        | Moss          | Sedge | Bulrush | Control | Moss          | Sedge | Bulrush | Control |
| Strata            | Moss                   | 29.91         | 0.06  | 0.19    | 0.27    | 0.14          | 0.13  | 0.06    | 0.41    |
|                   | Vascular               | 24.91         | 42.29 | 12.90   | 9.51    | 16.93         | 22.31 | 8.34    | 10.42   |
| Implanted species | Campyllum stellatum    | 20.98         | .     | .       | 0.01    | .             | .     | 0.45    | .       |
|                   | Scorpidium scorpioides | 2.92          | .     | .       | .       | .             | .     | .       | .       |
|                   | Carex lasiocarpa       | 0.06          | 0.35  | .       | .       | 0.01          | 0.31  | .       | .       |
|                   | Carex lacustris        | 0.08          | 2.50  | .       | .       | 0.13          | 0.91  | 0.09    | 0.02    |
|                   | Carex aquatilis        | 1.67          | 18.67 | .       | .       | 1.42          | 8.95  | 0.20    | 0.14    |
|                   | Carex utriculata       | 1.78          | 19.33 | 0.20    | 0.33    | 4.55          | 7.60  | 0.50    | 2.41    |
|                   | Scirpus validus        | 0.25          | 0.01  | 7.97    | 0.03    | 0.13          | 0.01  | 2.37    | 0.24    |
|                   |                        |               |       |         |         |               |       |         |         |

(continued on next page)

Table A1 (continued)

|                     |                                                | Within hollow |       |         |         | Around hollow |       |         |         |
|---------------------|------------------------------------------------|---------------|-------|---------|---------|---------------|-------|---------|---------|
|                     |                                                | Moss          | Sedge | Bulrush | Control | Moss          | Sedge | Bulrush | Control |
| Spontaneous species | <i>Abies balsamea</i> <sup>G</sup>             | 0.01          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Alnus rugosa</i> <sup>F</sup>               | .             | .     | .       | .       | 0.22          | .     | .       | .       |
|                     | <i>Alnus sp.</i> <sup>F</sup>                  | .             | 0.10  | .       | .       | .             | .     | .       | .       |
|                     | <i>Andromeda glaucophylla</i> <sup>F</sup>     | 0.28          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Aster nemoralis</i> <sup>F</sup>            | 0.02          | .     | .       | .       | 0.07          | .     | .       | .       |
|                     | <i>Aster novi-belgii</i> <sup>F</sup>          | 0.01          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Aster umbellatus</i> <sup>G</sup>           | .             | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Bidens cernua</i> <sup>G</sup>              | .             | .     | 0.03    | 0.01    | .             | .     | 0.05    | .       |
|                     | <i>Bidens frondosa</i> <sup>G</sup>            | .             | .     | 0.01    | 0.02    | .             | .     | .       | .       |
|                     | <i>Bidens sp.</i> <sup>G</sup>                 | 0.11          | 0.02  | 0.01    | 0.04    | .             | .     | .       | .       |
|                     | <i>Bryum pseudotriquetrum</i> <sup>F</sup>     | .             | 0.01  | 0.15    | 0.07    | 0.03          | 0.10  | 0.52    | 0.17    |
|                     | <i>Calamagrostis canadense</i> <sup>F</sup>    | 0.92          | 0.23  | 0.07    | 0.27    | 0.11          | 0.13  | 0.11    | 0.24    |
|                     | <i>Carex canescens</i> <sup>F</sup>            | 1.17          | .     | 0.11    | 0.07    | 0.50          | .     | 0.09    | 0.05    |
|                     | <i>Carex diandra</i> <sup>F</sup>              | .             | .     | 0.02    | .       | .             | .     | .       | .       |
|                     | <i>Carex exilis</i> <sup>F</sup>               | 0.15          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Carex flava</i> <sup>F</sup>                | .             | .     | 0.02    | .       | 0.07          | .     | 0.03    | .       |
|                     | <i>Carex livida</i> <sup>F</sup>               | 0.91          | 0.01  | 0.01    | .       | 0.06          | 0.01  | .       | .       |
|                     | <i>Carex sp.</i>                               | 0.03          | .     | 0.04    | .       | .             | 0.04  | .       | .       |
|                     | <i>Chamaedaphne calyculata</i> <sup>F</sup>    | 0.01          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Conioselinum chinense</i> <sup>F</sup>      | .             | .     | 0.01    | .       | .             | .     | .       | .       |
|                     | <i>Drosera intermedia</i> <sup>F</sup>         | 0.55          | .     | 0.01    | .       | 0.03          | .     | .       | .       |
|                     | <i>Drosera rotundifolia</i> <sup>F</sup>       | 0.22          | 0.06  | 0.04    | 0.13    | 0.01          | 0.01  | .       | .       |
|                     | <i>Epilobium palustre</i> <sup>F</sup>         | 0.14          | .     | 0.06    | .       | .             | .     | .       | .       |
|                     | <i>Equisetum arvense</i> <sup>G</sup>          | 0.02          | 0.06  | 0.26    | 0.94    | 0.12          | 0.08  | 0.05    | 0.71    |
|                     | <i>Eriophorum angustifolium</i> <sup>F</sup>   | 0.17          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Eriophorum viridicarinatum</i> <sup>F</sup> | 0.09          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Eupatorium maculatum</i> <sup>G</sup>       | .             | .     | .       | .       | .             | .     | 0.05    | .       |
|                     | <i>Galium trifidum</i> <sup>F</sup>            | 0.01          | .     | 0.01    | 0.01    | .             | .     | .       | .       |
|                     | <i>Hypericum virginicum</i> <sup>F</sup>       | 0.30          | 0.17  | 0.24    | 0.20    | 0.01          | 0.01  | 0.02    | .       |
|                     | <i>Iris versicolor</i> <sup>F</sup>            | 0.02          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Juncus brevicodatus</i> <sup>F</sup>        | 2.48          | 0.40  | 0.80    | 0.85    | 0.87          | 0.69  | 0.70    | 0.18    |
|                     | <i>Juncus effusus</i> <sup>F</sup>             | 0.16          | 0.01  | 0.04    | 0.01    | 0.67          | 1.10  | 0.23    | 0.12    |
|                     | <i>Kalmia polifolia</i> <sup>F</sup>           | 0.04          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Larix laricina</i> <sup>F</sup>             | .             | 0.01  | .       | .       | .             | .     | .       | .       |
|                     | <i>Lemna minor</i> <sup>G</sup>                | .             | .     | .       | 0.02    | .             | .     | .       | .       |
|                     | <i>Lycopodium uniflorum</i> <sup>F</sup>       | 1.81          | 0.45  | 0.30    | 0.32    | 0.63          | 0.15  | 0.02    | 0.14    |
|                     | <i>Menyanthes trifoliata</i> <sup>F</sup>      | 0.06          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Myrica gale</i> <sup>F</sup>                | 0.36          | 0.01  | 0.01    | 0.05    | 0.05          | 0.04  | 0.05    | 0.08    |
|                     | <i>Onoclea sensibilis</i> <sup>G</sup>         | 0.04          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Pleurosium schreberi</i> <sup>F</sup>       | 0.01          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Potentilla palustris</i> <sup>F</sup>       | 0.01          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Rhynchospora alba</i> <sup>F</sup>          | 2.44          | 0.01  | 0.06    | 0.01    | 0.02          | .     | .       | .       |
|                     | <i>Salix candida</i> <sup>F</sup>              | 0.02          | .     | .       | .       | 0.02          | 0.01  | .       | .       |
|                     | <i>Salix sp.</i>                               | 0.15          | 0.20  | 0.13    | 0.26    | 0.12          | 0.13  | 0.20    | 0.05    |
|                     | <i>Salvinia sp.</i>                            | .             | 0.02  | 0.01    | .       | .             | .     | .       | .       |
|                     | <i>Sanguisorba canadensis</i> <sup>F</sup>     | 0.05          | 0.03  | 0.01    | 0.06    | .             | 0.01  | 0.05    | .       |
|                     | <i>Sarracenia purpurea</i> <sup>F</sup>        | 0.26          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Scirpus cyperinus</i> <sup>F</sup>          | 2.69          | 0.10  | 1.00    | 3.81    | 3.28          | 0.44  | 0.80    | 1.98    |
|                     | <i>Solidago graminifolia</i> <sup>G</sup>      | 0.01          | 0.01  | 0.10    | .       | .             | .     | 0.05    | 0.02    |
|                     | <i>Solidago sp.</i>                            | .             | 0.03  | .       | 0.02    | .             | .     | .       | .       |
|                     | <i>Solidago uliginosa</i> <sup>F</sup>         | 0.01          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Sphagnum sp.</i>                            | 0.13          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Spiraea latifolia</i> <sup>F</sup>          | 0.08          | 0.07  | 0.05    | 0.14    | 0.02          | 0.04  | 0.17    | 0.36    |
|                     | <i>Trichophorum cespitosum</i> <sup>F</sup>    | .             | .     | .       | .       | .             | 0.01  | .       | .       |
|                     | <i>Trichophorum alpinum</i> <sup>F</sup>       | 0.05          | 0.01  | 0.06    | .       | 0.03          | .     | 0.10    | 0.02    |
|                     | <i>Triglochin maritima</i> <sup>F</sup>        | 0.01          | .     | .       | .       | 0.03          | .     | .       | .       |
|                     | <i>Triglochin palustris</i> <sup>F</sup>       | 0.22          | 0.04  | 0.08    | 0.17    | 0.30          | .     | 0.14    | .       |
|                     | <i>Typha latifolia</i> <sup>G</sup>            | 2.25          | 0.36  | 0.83    | 2.19    | 1.15          | 1.08  | 1.05    | 1.33    |
|                     | <i>Utricularia intermedia</i> <sup>F</sup>     | 1.83          | 0.08  | 0.61    | 0.20    | 0.18          | 0.08  | 0.08    | 0.41    |
|                     | <i>Vaccinium oxycoccus</i> <sup>F</sup>        | 0.20          | .     | .       | .       | .             | .     | .       | .       |
|                     | <i>Viola sp.</i>                               | 0.03          | .     | 0.05    | 0.02    | .             | .     | .       | .       |

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